AMR RAFIK AHMED ABDELMOATY

COMPUTER ENGINEERING GRADUATE | MACHINE LEARNING & DATA INTELLIGENCE

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PROFESSIONAL PROFILE

2026 Computer Engineering graduate with project experience in machine learning, agent-based AI, and data engineering. Developed a telecom intelligence assistant combining LangGraph workflows, RAG, SQL-backed customer lookup, and a tuned churn prediction model. Completed internships involving Oracle SQL, Informatica PowerCenter, SSIS, Power BI, and machine learning evaluation, alongside contract work assessing AI-generated code and function-calling behavior.

EDUCATION

B.Sc. in Computer Engineering | Benha University, Shoubra Faculty of Engineering | Jun 2026

GPA: 3.18

TECHNICAL SKILLS

Programming & analytics: Python, Pandas, NumPy, SQL (Oracle SQL, SQLite), exploratory data analysis, Power BI.

Machine learning & computer vision: Scikit-learn, Random Forest, GridSearchCV, cross-validation, TensorFlow, Keras, OpenCV, CNNs, preprocessing, model evaluation.

Generative & Agentic AI: Gemini API, LangGraph, agent workflows, RAG, FAISS, embeddings, tool/function calling, prompt design, LLM evaluation.

Data & application engineering: Informatica PowerCenter, SSIS, SSMS, ETL, data warehousing, FastAPI, REST APIs, Streamlit, Pydantic, Git, Pytest, structured logging.

FLAGSHIP AI & MACHINE LEARNING PROJECT

Telecom Customer Intelligence AI Assistant | Independent AI/ML Project | Hosted Offline Demo

- Developed a telecom assistant with two specialized LangGraph agent branches, FAISS-based RAG retrieval, and parameterized SQLite customer lookups; integrated Gemini routing and tool calling in local live mode.
- Trained and tuned a Scikit-learn Random Forest using GridSearchCV and grouped cross-validation, achieving F1 = 0.856 and ROC-AUC = 0.979 on 635 held-out UCI Iranian Churn records.
- Implemented validated tool calling, FastAPI REST endpoints, Streamlit, and structured request logging. Documented 143 passing automated tests, 30/30 offline evaluation cases, and limited live Gemini verification; published a separate offline-only cloud demo.

RELEVANT DATA & MACHINE LEARNING INTERNSHIPS

Data Integration Intern | EJADA Systems, Cairo | Jul 2025 – Aug 2025

- Collaborated with two teammates on a Banking Analytics System using Oracle SQL and Informatica PowerCenter, with a 13-table relational schema covering accounts, transactions, loans, investments, and other banking data.
- Worked with analytical SQL involving JOINs, aggregations, CTEs, and ranking functions, alongside Informatica ETL mappings for banking analytics.

Machine Learning Intern | National Telecommunication Institute (NTI), Cairo | Jul 2025 – Aug 2025

- Prepared image and tabular datasets through cleaning, preprocessing, and augmentation for classification tasks.
- Evaluated classification models using precision, recall, and ROC-AUC.

Data Analysis Intern | EJADA Systems, Cairo | Jul 2024 – Aug 2024

- Worked on an end-to-end AdventureWorks business intelligence project using SSMS, SSIS, and Power BI to integrate customer, product, and sales data for warehouse-based analysis.
- Imported, cleaned, and transformed data and developed Power BI reports and visualizations to support sales analysis.

AI MODEL EVALUATION & ANNOTATION EXPERIENCE

AI Code Evaluator | DataAnnotation, Independent Contractor | Jul 2026 – Present

- Evaluate AI-generated code for correctness, reasoning, edge cases, and instruction adherence; document comparative technical judgments.

AI Video Annotator | micro1, Remote Contractor | Jun 2026 – Present

- Annotate video data for AI model training under task-specific labeling and consistency guidelines.

AI Training Specialist & Quality Reviewer | Alignerr, Remote Contractor | Sep 2025 – Jun 2026

- Designed spoken prompts to elicit AI function calls and evaluated model responses and voice output against task requirements.
- Created coding prompts, compared AI-generated code and reasoning, and completed VQA/RLHF review, transcription, and OCR-to-JSON tasks.

ADDITIONAL PROJECTS

Hayah — Smart Assistive Wearable Glasses | Graduation Project, Grade A+

- Developed an embedded AI wearable integrating real-time facial emotion recognition, gesture detection, and visual scene analysis; prepared annotated image/video datasets.

LEADERSHIP & ACHIEVEMENTS

2nd Place Globally | Shell Eco-marathon Autonomous Programming Competition | 2025

- Led Shoubra Racing Team’s autonomous control, decision-making, and simulation testing for a self-driving electric vehicle competing against teams from 20+ universities worldwide.